

IMO 2026 — Problem 2

Solution by xAI Grok 4.5

2026-07-21

Source: `results/grok-4.5/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC , respectively. Let points K and L be chosen inside triangles BMC and BNC , respectively, such that K lies inside the angle LBA , L lies inside the angle ACK , and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL . Prove that $OM = ON$.

Solution document

Status

partial

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Approaches tried

- Numerics: claim true ($OM=ON$ for the 1-param family).
- Isosceles case: proved by reflection.
- Spiral sims, sine law expressions known.
- Algebraic target with $A=0$ known.

Current best

Working toward a geometric proof: perhaps by showing that the reflection of O across the midperp of MN coincides with O , or that triangle OMN is isosceles with $OM = ON$ by some equal angles or congruent triangles. Or use Cartesian with origin at mid of MN .

Full proof

(not yet)